FATEC

Curso de Tecnologia em Análise e Desenvolvimento de Sistemas

IAC-001
Arquitetura e Organização de Computadores

Tema: Lógica Digital

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Tabela Verdade

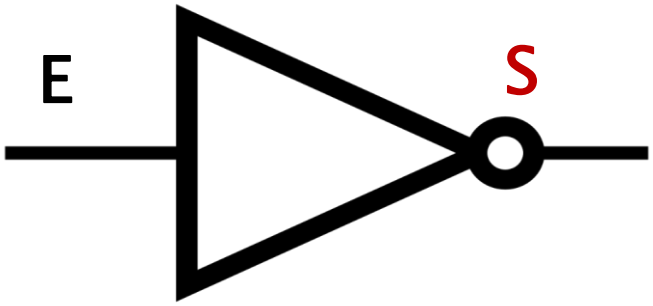
- Serve para mostrar **todas possibilidades** que a **entrada** pode assumir e o valor associado da **saída** para **cada uma dessas entradas**
- Tem tantas linhas quanto a quantidade de combinações diferentes que as entradas possam assumir

- Exemplos:
- uma entrada => 2 linhas
- 2 entradas => 4 linhas
- 3 entradas => 8 linhas
- 4 entradas = > 16 linhas
- 5 entradas => 32 linhas
- 6 entradas => 64 linhas

A	S
0	
1	

A	B	S
0	0	
0	1	
1	0	
1	1	

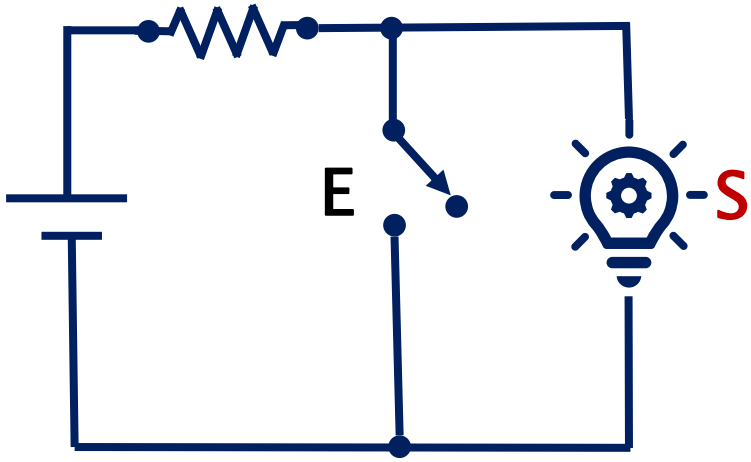
Porta NÃO (NOT) = INVERSORA (INVERTER)



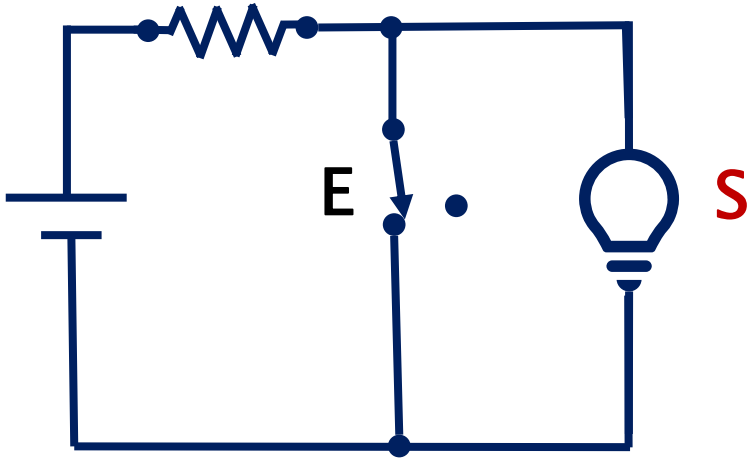
- Só uma entrada e uma saída
- Entrada pode ser 0 ou 1
- Saída é o inverso da entrada

- Tabela Verdade da porta Inversora (NOT)

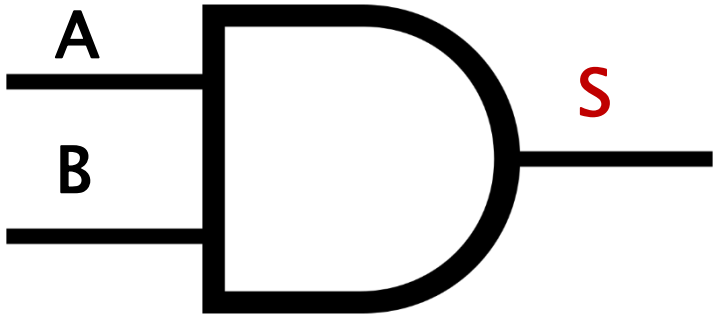
E	S
0	1
1	0



Quando o interruptor é ligado a lâmpada se apaga

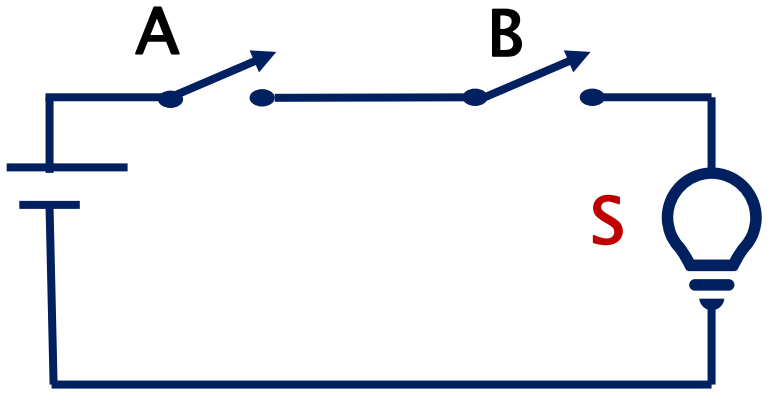


Porta E (AND)

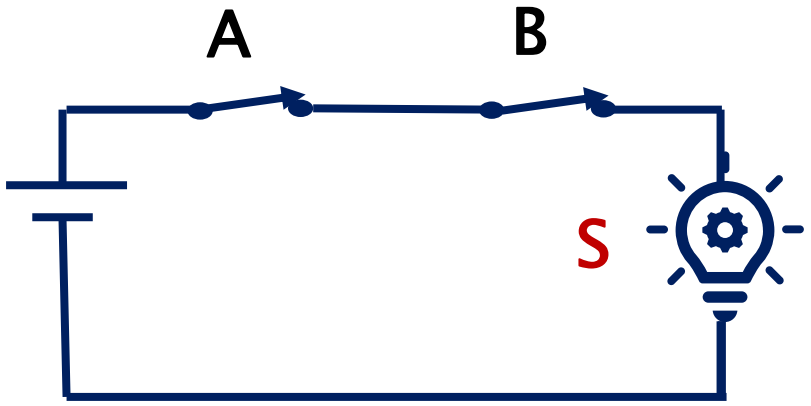


- Tabela Verdade

A	B	S
0	0	0
0	1	0
1	0	0
1	1	1



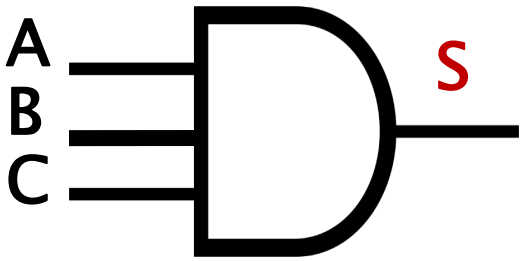
Só acende quando todas chaves ligadas

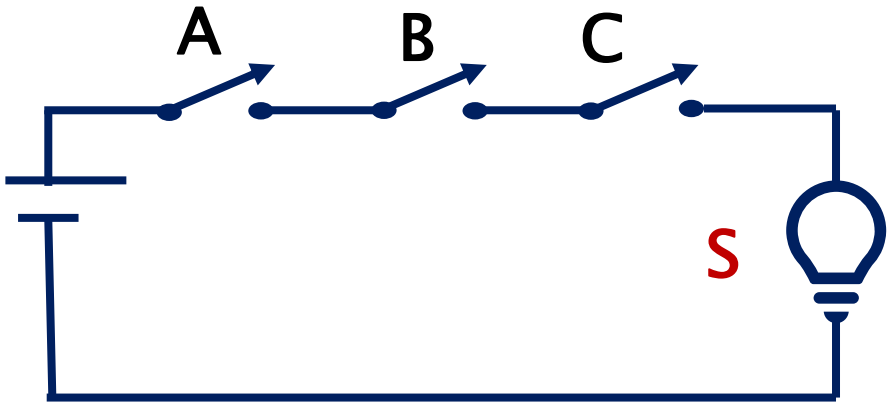


Porta E (AND) – 3 Entradas

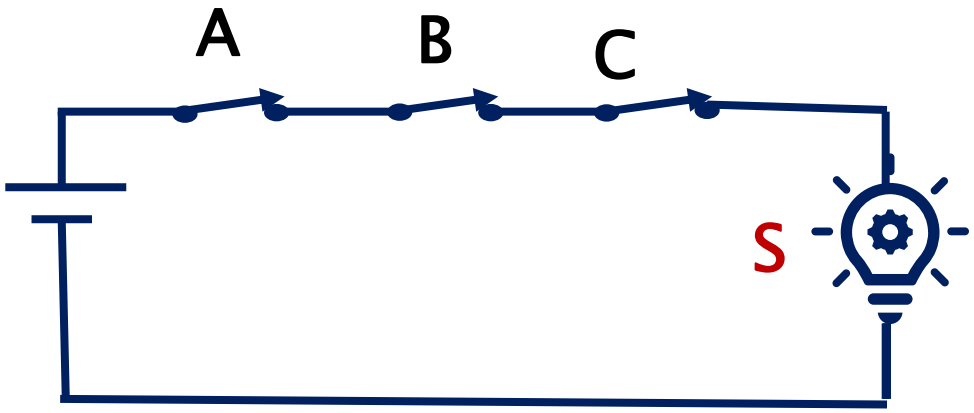
- Tabela Verdade

A	B	C	S
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



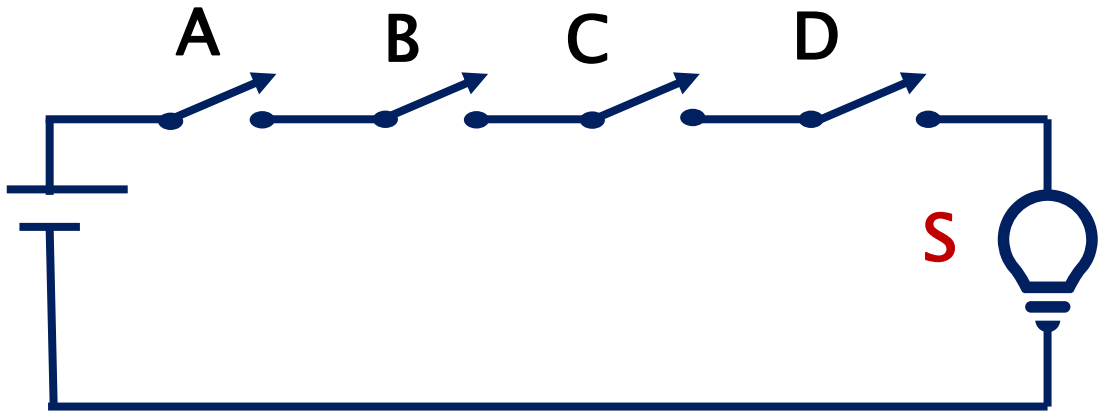


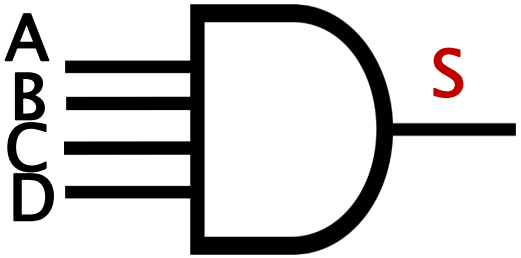
Só acende quando todas chaves ligadas

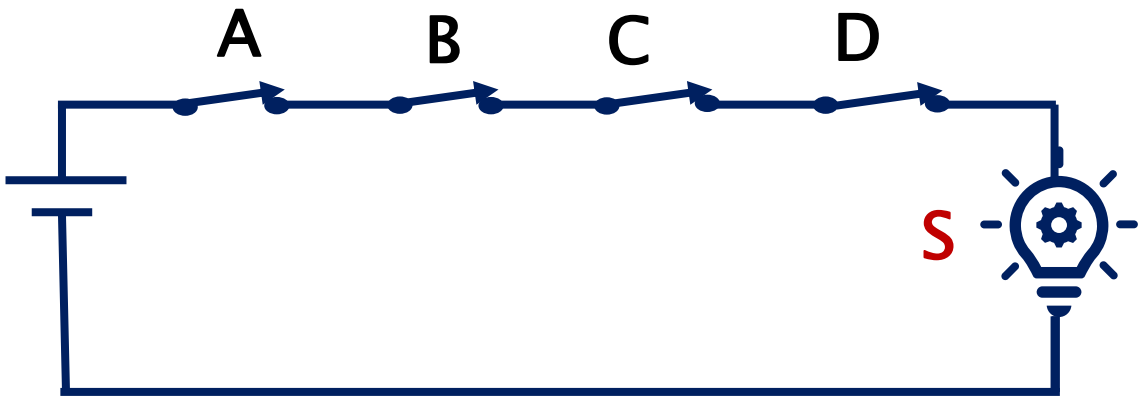


Porta E (AND) – 4 Entradas – Exercício

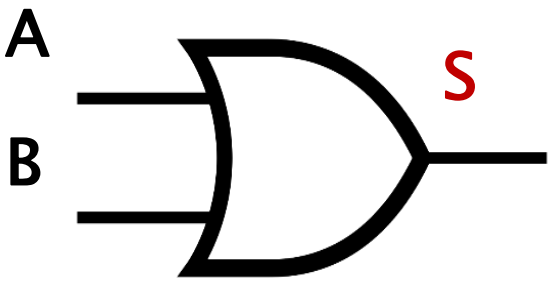
- Tabela Verdade
- Quantas entradas?
- Quantas possibilidades?
- Quantas linhas ?
- Quantas Colunas ?

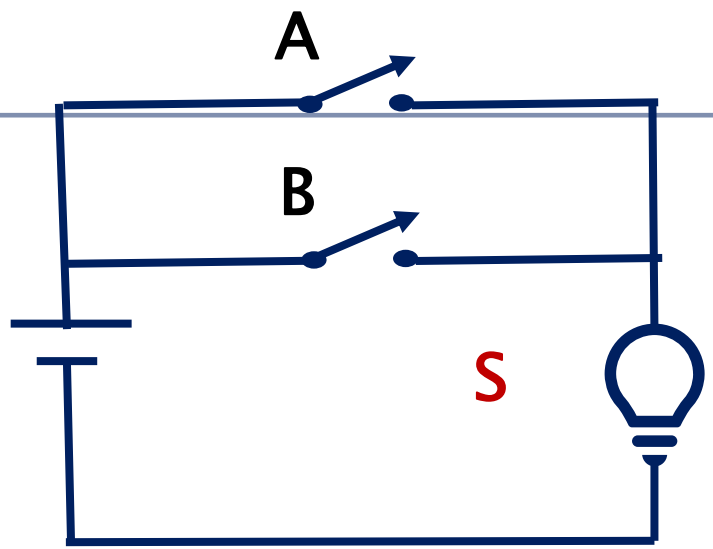






Porta OU (OR)

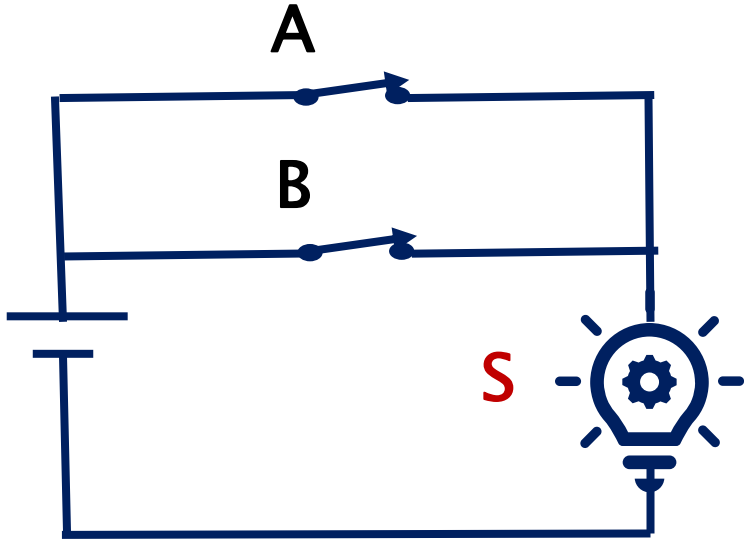




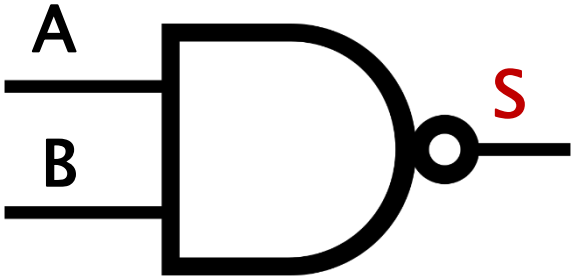
- Tabela Verdade

A	B	S
0	0	0
0	1	1
1	0	1
1	1	1

Acende com PELO MENOS uma chave ligada



Porta NÃO E (NAND)

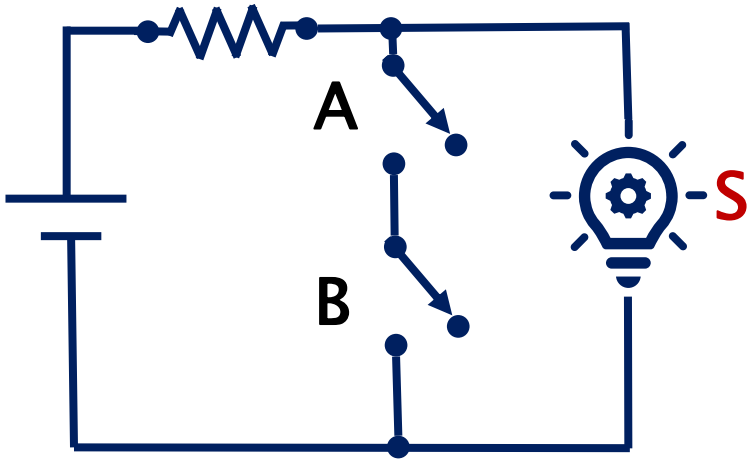


- Tabela Verdade

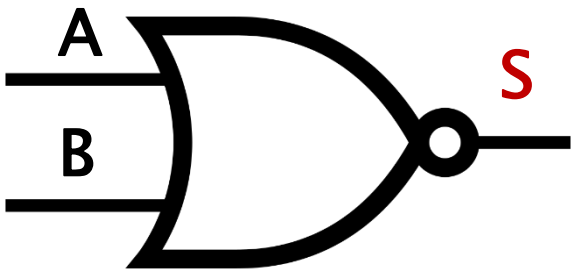
A	B	S
0	0	1
0	1	1
1	0	1
1	1	0

- É a negação da porta E (AND)

$$S = \overline{A.B}$$



Porta NÃO OU (NOR)

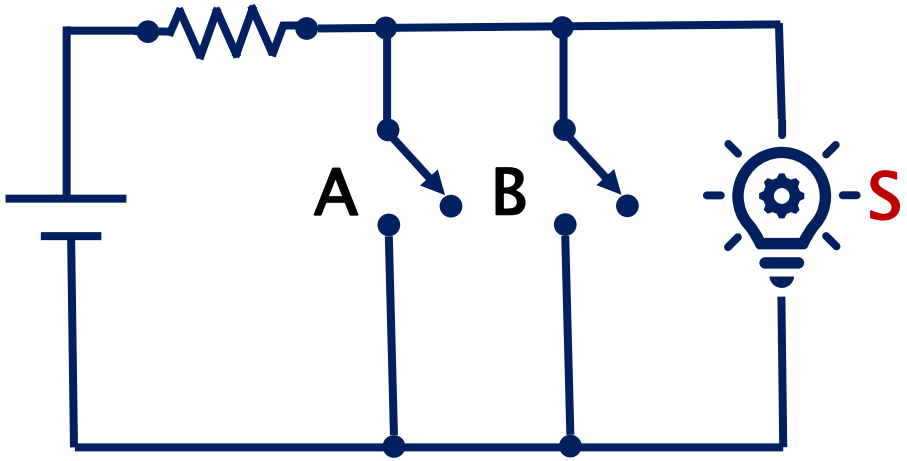


- Tabela Verdade

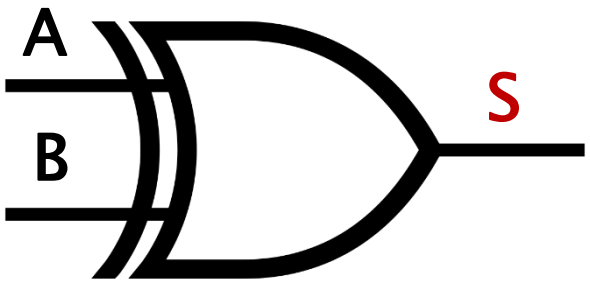
A	B	S
0	0	1
0	1	0
1	0	0
1	1	0

- É a negação da porta OU (OR)

$$S = \overline{A + B}$$



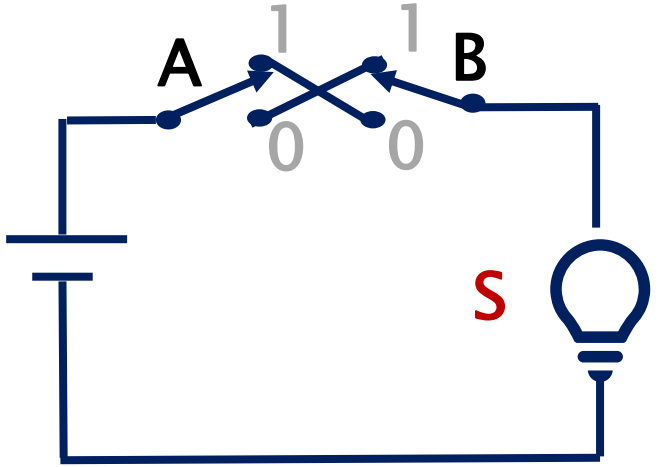
Porta OU EXCLUSIVO (XOR) – SOMADOR

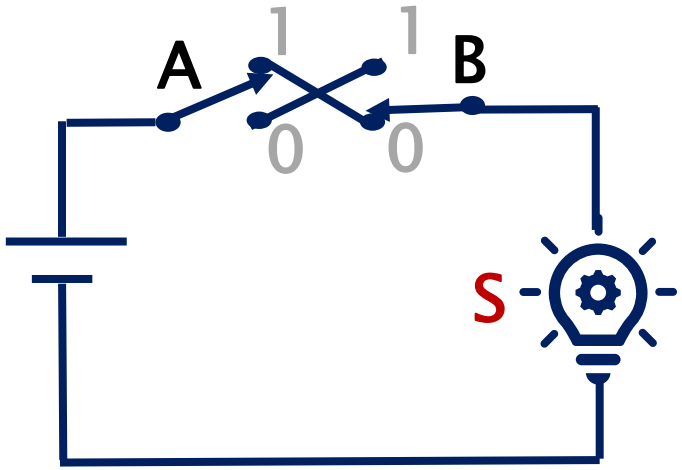


- Tabela Verdade

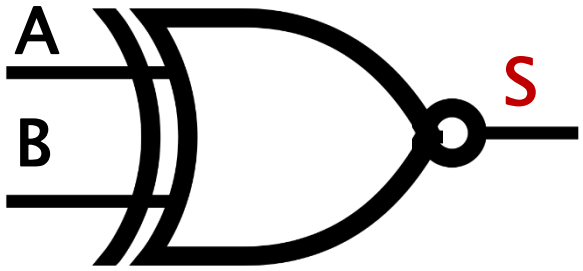
A	B	S
0	0	0
0	1	1
1	0	1
1	1	0

$$S = A \oplus B$$





NÃO OU EXCLUSIVO (Exclusive NOR, XNOR)



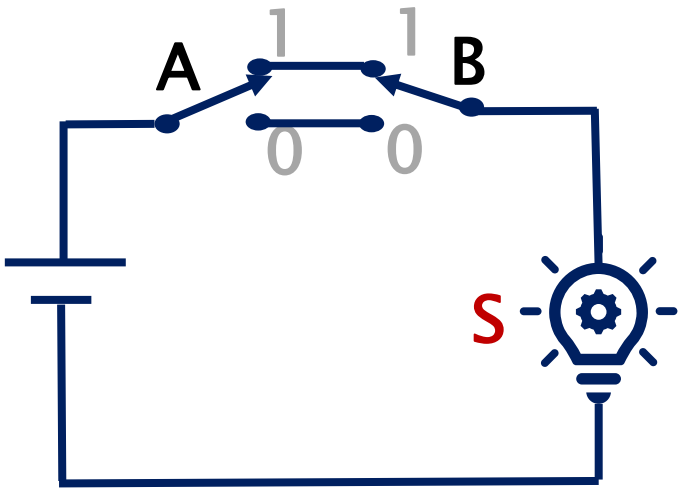
- Tabela Verdade

A	B	S
0	0	1
0	1	0
1	0	0
1	1	1

- **Comparador** de **igualdade**

$$S = \overline{A \oplus B} = \overline{A} \oplus B$$

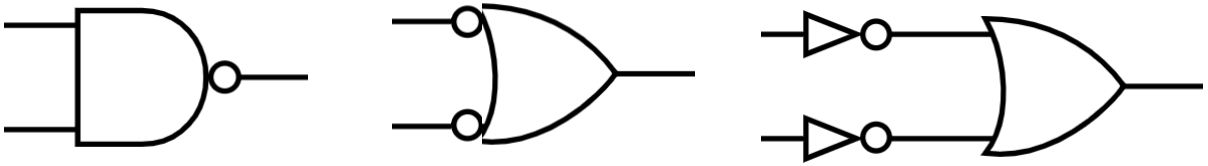
- Interruptor paralelo ("*three-Way*")



Teorema de DeMorgan (Relacionamento entre Portas Lógicas)

$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

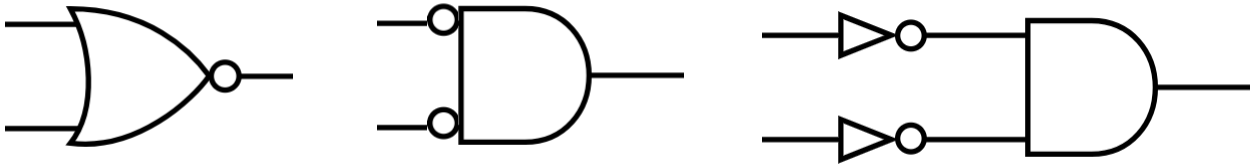
São equivalentes para a porta NAND (Não E):



A	B	S
0	0	1
0	1	1
1	0	1
1	1	0

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

São equivalentes para a porta NOR (Não OU):

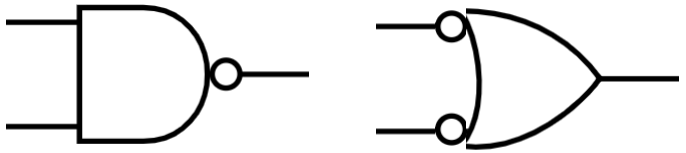


A	B	S
0	0	1
0	1	0
1	0	0
1	1	0

Teorema de DeMorgan

$$\overline{A \,.\, B} = \overline{A} + \overline{B}$$

São equivalentes:



Equivalência para a porta **NAND** (Não E)

A	B	$\overline{A}$	$\overline{B}$	$\overline{A} \cdot \overline{B}$	$\overline{A} + \overline{B}$	A.B	$\overline{A \,.\, B}$	S
0	0	1	1	1	1	0	1	1
0	1	1	0	0	1	0	1	1
1	0	0	1	0	1	0	1	1
1	1	0	0	0	0	1	0	0

São equivalentes

$$\overline{A \,.\, B} = \overline{A} + \overline{B}$$

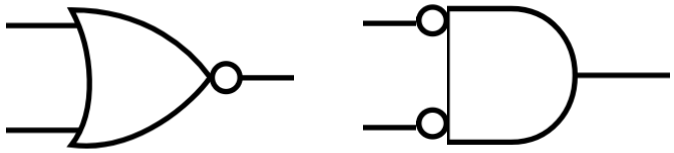
A	B	S
0	0	1
0	1	1
1	0	1
1	1	0

Teorema de DeMorgan

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

Equivalência para a porta **NOR** (Não OU)

São equivalentes:



A	B	$\overline{A}$	$\overline{B}$	$\overline{A} \cdot \overline{B}$	A+B	$\overline{A + B}$	S
0	0	1	1	1	0	1	1
0	1	1	0	0	1	0	0
1	0	0	1	0	1	0	0
1	1	0	0	0	1	0	0

São equivalentes

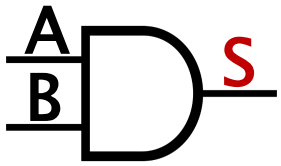
$$\overline{A + B} = \overline{A}.\overline{B}$$

A	B	S
0	0	1
0	1	0
1	0	0
1	1	0

Resumo de Portas Lógicas

AND (E)

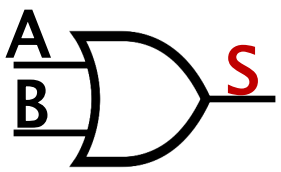
S=A.B



A	B	S
0	0	0
0	1	0
1	0	0
1	1	1

OR (OU)

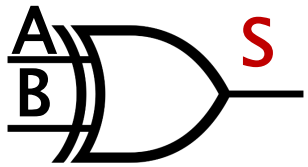
S=A+B



A	B	S
0	0	0
0	1	1
1	0	1
1	1	1

XOR Somador

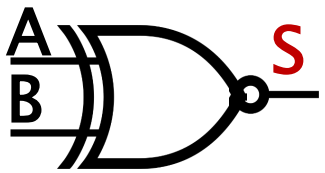
$S = A \oplus B$



A	B	S
0	0	0
0	1	1
1	0	1
1	1	0

NXOR Comparador

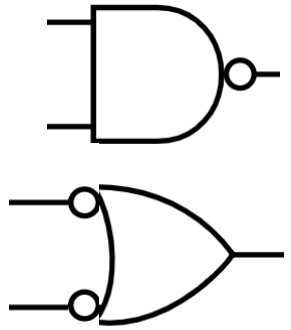
$S = A \overline{\oplus} B = \overline{A \oplus B}$



A	B	S
0	0	1
0	1	0
1	0	0
1	1	1

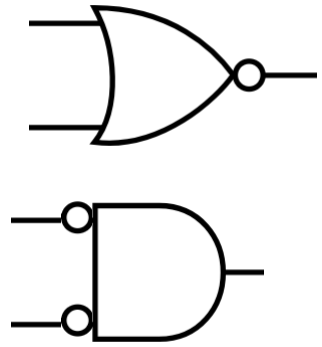
DeMorgan

$\overline{A \,.\, B} = \overline{A} + \overline{B}$



DeMorgan

$\overline{A + B} = \overline{A} \cdot \overline{B}$



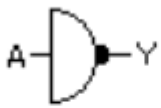
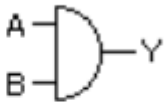
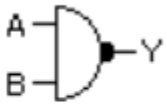
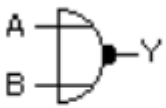
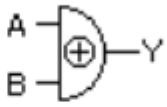
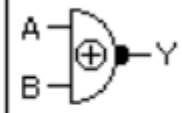
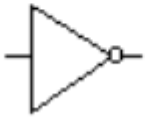
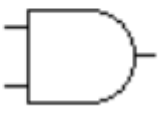
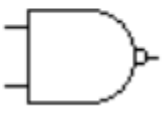
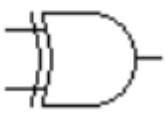
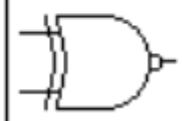
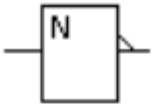
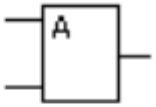
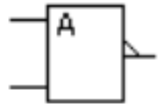
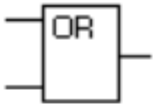
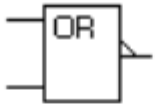
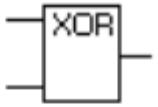
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Bibliografia

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Diferentes Formas de Representação

	INVERT	AND	NAND	OR	NOR	EX - OR	EX - NOR
European							
American							
IBM ALD's							
Boolean	$Y = \overline{A}$	$Y = A \cdot B$	$Y = \overline{A \cdot B}$	$Y = A + B$	$Y = \overline{A + B}$	$Y = A + B$	$Y = \overline{A + B}$
Truth Table	A Y L H H L	A B Y L L L H L L L H L H H H	A B Y L L H H L H L H H H H L	A B Y L L L H L H L H H H H H	A B Y L L H H L L L H L H H L	A B Y L L L H L H L H H H H L	A B Y L L H H L L L H L H H H

Álgebra de Boole (álgebra Booleana)

- Para saber mais:
  - 8Set25
  - https://www.electronics-tutorials.ws/boolean/boolean-algebra-simplification.html
-
- A Álgebra Booleana é usada para simplificar combinações lógicas sem usar tabela verdade (que ppode conter muitas linhas e ser morosa para ser construída)

- É usada para diminuir a quantidade de circuitos *(“chips”)* dentro de uma placa.
- Também serve para deixar os alunos loucos.
  - Só por isso não vamos cobrar isso; tá doido sô!